

Quantifiers

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Review

A **proposition** is a statement with exactly one truth value: **T** or **F**.
For propositions P and Q , the implication $P \Rightarrow Q$ is false exactly when P is true and Q is false.
We now consider mathematical sentences whose truth values depend on variables such as x and n .

Divisibility and prime numbers

Let d and n be integers.

We say that d **divides** n , written $d \mid n$, if $n = dk$ for some integer k .

An integer $n > 1$ is **prime** if its only positive divisors are 1 and n .

An integer $n > 1$ is **composite** if it is not prime.

Testing whether an integer is prime

```
def check_prime(n):  
    if n <= 1:  
        return False  
    i = 2  
    while i * i <= n:  
        if i divides n:  
            return False  
        i = i + 1  
    return True
```

The algorithm tests possible divisors only while $i^2 \leq n$.

The question is not whether it works on several examples, but whether it returns the correct answer for every integer input.

The correctness claim

The algorithm is correct if the following statement is true:

For every integer n , `check_prime(n)` returns **T** exactly when n is prime.

The statement refers to an arbitrary input n .

To write it symbolically, we need

- ▶ statements whose truth values depend on n , and
- ▶ notation for “for every integer n .”

Open sentences

An **open sentence** contains a variable and does not have a single truth value until the variable is specified.

For example,

$$x > 0$$

is true for some values of x and false for others.

Substituting a value produces a proposition:

$$3 > 0 \text{ is } \mathbf{T}, \quad -2 > 0 \text{ is } \mathbf{F}.$$

Universe

A **set** is a collection of objects. A **universe** is the set of values that a variable is allowed to take.

The notation $x \in U$ means that x is an element of the set U .

We write

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

for the set of all integers.

The universe is part of the meaning of a sentence containing a variable.

Predicates

Let U be a universe and let $x \in U$.

A **predicate** $P(x)$ is an open sentence whose truth value depends on x .

For $x \in \mathbb{Z}$, define

$$P(x) = "x > 0."$$

The expression $P(x)$ is not a proposition while x remains unspecified.

Substituting a value

Continue with the predicate

$$P(x) = "x > 0", \quad x \in \mathbb{Z}.$$

Substituting $x = 3$ gives the true proposition $P(3)$.

Substituting $x = -2$ gives the false proposition $P(-2)$.

A **variable assignment** selects a value from the universe for a variable.

Assigning every variable in a predicate produces a proposition.

More than one variable

A predicate may contain several variables.

For $x, y \in \mathbb{Z}$, define

$$R(x, y) = "x \leq y."$$

Assigning both variables produces propositions:

$$R(2, 5) \text{ is } \mathbf{T}, \quad R(5, 2) \text{ is } \mathbf{F}.$$

If either variable remains unspecified, $R(x, y)$ remains a predicate.

Quantifiers

A **quantifier** specifies how a variable ranges over its universe.
The two basic quantifiers are:

name	symbol	read as
universal quantifier	$\forall$	for every
existential quantifier	$\exists$	there exists

A variable controlled by a quantifier is **bound**. A variable that is neither assigned nor bound is **free**.

A statement with no free variables is a proposition.

Universal quantifier

Let

$$U = \{-2, -1, 0, 1, 2\}, \quad N(x) = "x^2 \geq 0."$$

The proposition

$$(\forall x \in U)N(x)$$

is read "for every x in U , $N(x)$ is true."

It is true because $N(a)$ is true for every element $a \in U$.

Counterexamples

Let

$$U = \{-2, -1, 0, 1, 2\}, \quad P(x) = "x > 0."$$

The proposition $(\forall x \in U)P(x)$ is false because $P(0)$ is false.

A **counterexample** to $(\forall x \in U)P(x)$ is an element $a \in U$ for which $P(a)$ is false.

One counterexample is enough to disprove a universal proposition.

Existential quantifier

Continue with

$$U = \{-2, -1, 0, 1, 2\}, \quad P(x) = "x > 0."$$

The proposition

$$(\exists x \in U)P(x)$$

is read "there exists an x in U for which $P(x)$ is true."

A **witness** is a value that makes an existential proposition true. Here 1 is a witness because $P(1)$ is true.

No witness

Let

$$U = \{-2, -1, 0, 1, 2\}, \quad S(x) = "x^2 < 0."$$

The proposition $(\exists x \in U)S(x)$ is false.

The predicate is false for every element of U , so no witness exists.

To disprove an existential proposition, every possible value must be ruled out.

Finite universes

Suppose the universe contains exactly

$$U = \{a_1, a_2, \dots, a_n\}.$$

Universal quantification is a conjunction:

$$(\forall x \in U)P(x) \equiv P(a_1) \wedge \dots \wedge P(a_n).$$

Existential quantification is a disjunction:

$$(\exists x \in U)P(x) \equiv P(a_1) \vee \dots \vee P(a_n).$$

Predicates for the algorithm

Let the universe be $\mathbb{Z}$. Define

$$C(n) = \text{"check_prime}(n) \text{ returns } \mathbf{T}"$$

and

$$P(n) = \text{"}n \text{ is prime."}$$

For a particular integer, such as 17, the expressions $C(17)$ and $P(17)$ are propositions.

The expressions $C(n)$ and $P(n)$ are predicates while n is free.

Correctness as a proposition

For each integer n , the biconditional

$$C(n) \Leftrightarrow P(n)$$

says that the algorithm's answer agrees with whether n is prime.
The complete correctness claim is

$$(\forall n \in \mathbb{Z})(C(n) \Leftrightarrow P(n)).$$

This is a proposition because n is bound by $\forall$.

An integer for which $C(n)$ and $P(n)$ have different truth values would be a counterexample.

Universal implications

Let $P(x)$ and $Q(x)$ be predicates on the same universe U .

The expression $P(x) \Rightarrow Q(x)$ is a predicate while x is free.

The proposition

$$(\forall x \in U)(P(x) \Rightarrow Q(x))$$

is false exactly when some $a \in U$ makes $P(a)$ true and $Q(a)$ false.

Such an element a is a counterexample to the universal implication.

Vacuous truth

Suppose no element of U makes the predicate $P(x)$ true.

Then $P(a)$ is false for every $a \in U$.

By the truth table for implication, $P(a) \Rightarrow Q(a)$ is true for every $a \in U$, regardless of $Q(a)$.

Therefore,

$$(\forall x \in U)(P(x) \Rightarrow Q(x))$$

is vacuously true.

Summary

- ▶ A predicate becomes a proposition after every variable is assigned or bound.
- ▶ The universe specifies the values over which a variable ranges.
- ▶ The quantifiers $\forall$ and $\exists$ express “every” and “there exists.”
- ▶ A counterexample refutes a universal proposition; a witness establishes an existential proposition.
- ▶ Algorithm correctness can be stated as a quantified proposition.